

Dr Michèle Christopher Renard

mrenard@nus.edu.sg | Singapore

[LinkedIn](#) | [ORCID 0000-0003-4517-1316](#) | [Google Scholar](#) | [ResearchGate](#)

Environmental physiologist and quantitative researcher. I model how the built thermal environment affects sleep, physiological strain and health in real-world populations. At the National University of Singapore I lead the statistical analysis for two national research programmes, working across large longitudinal wearable and sensor datasets, randomised field interventions, and evidence synthesis that feeds national heat-health policy. PhD in Sport, Health and Exercise Sciences; eleven peer-reviewed publications; four PROSPERO-registered systematic reviews; three doctoral candidates and a postdoctoral researcher under supervision.

EXPERIENCE

Research Fellow — National University of Singapore, Yong Loo Lin School of Medicine

Human Potential Translational Research Programme & Heat Resilience and Performance Centre | Apr 2025 – Present

- Analytical lead across two NRF-funded national programmes — **Project HEATS** and **Cooling Singapore 2.0** — owning the statistical workflow from analysis plan to publication.
- Lead the flagship analysis of indoor heat exposure and sleep across **122 participants and 10,303 person-nights** (three field waves, Jun 2025 – Jul 2026), using mixed-effects models with Mundlak decomposition to separate behavioural adaptation from physiological strain, and spline and segmented methods to locate non-linear thresholds.
- Model laboratory heat-exposure responses across vulnerable population groups (children, unacclimatised adults, older adults) and translate outputs into policy recommendations shaping Singapore's national heat-health advisories.
- Lead four systematic reviews and supervise four early-career researchers; work featured in *The Straits Times* and *Lianhe Zaobao* following the NRF CREATE Symposium 2026.

Postdoctoral Research Associate — King's College London, Department of Nutritional Sciences

Jul 2024 – Apr 2025 (Research Assistant, Feb – Jul 2024)

- Led research on cooking and food behaviours during the UK cost-of-living crisis, from recruitment and data collection through to publication.
- Delivered mixed-methods analysis across large survey datasets — hierarchical regression, non-parametric group comparison, and thematic analysis of focus group and open-text data.
- Registered and led a scoping review on food and sustainability living laboratories in higher education.

Assistant Lecturer — Atlantic Technological University, Sligo, Ireland | Oct 2021 – Jun 2022

Module lead for MSc Advanced Performance Nutrition; lectured across undergraduate exercise physiology, anatomy and nutrition; supervised ten undergraduate research projects.

Health Advisor — Bupa, UK | Mar 2018 – Jul 2019

Delivered clinical health assessments: 12-lead ECG, spirometry, sub-maximal CPET, point-of-care blood analysis (trained phlebotomist), and motivational-interviewing-based lifestyle coaching.

TECHNICAL SKILLS

Statistical modelling — Linear mixed-effects models; Mundlak within/between decomposition; fixed effects with cluster-robust SEs; piecewise/segmented threshold models with AIC knot search; restricted cubic splines; GLMs and GEE; cluster bootstrap; multiplicity control.

Evidence synthesis — PRISMA 2020 / PRISMA-P; PROSPERO registration (4 records); MMAT; Cochrane RoB 2.0; meta-analysis appraisal, publication bias and certainty of evidence.

Programming — Python (statsmodels, pandas, seaborn, python-docx, PyMuPDF); R (tidyverse, ggplot2); SPSS; Git/GitHub; Bash; Quarto/HTML.

Physiological measurement — Wearable and in-home environmental monitoring; actigraphy and sleep outcomes (TST, WASO, efficiency); core and skin temperature; VO₂max and CPET; lactate threshold; RMR; anthropometry; 12-lead ECG; spirometry; phlebotomy.

SELECTED PUBLICATIONS

Eleven peer-reviewed publications; three under review; six in preparation. Full list on [ORCID](#).

1. George, C. V., Bell, Z., **Renard, M.**, et al. (2026). Pregnancy and postpartum nutritional wants and needs in athletes: A cross-sectional survey in the UK and Republic of Ireland. *Performance Nutrition*, **2**(1), 14.
2. **Renard, M.**, Bell, Z., Jamshidvand, M., et al. (2025). Domestic cooking and food behaviours during the COVID-19 pandemic and the cost-of-living crisis: A scoping review. *Appetite*, **216**, 108311.
3. **Renard, M.**, Kelly, D. T., Ní Chéilleachair, N., Lavelle, F., & Ó Catháin, C. (2023). Cooking and food skills confidence of team sport athletes in Ireland. *Nutrition Bulletin*, **48**(3), 329–342.
4. Taim, B. C., Ó Catháin, C., **Renard, M.**, et al. (2023). The prevalence of menstrual cycle disorders and menstrual cycle-related symptoms in female athletes: A systematic literature review. *Sports Medicine*, **53**(10), 1963–1984.
5. **Renard, M.**, Kelly, D. T., Ní Chéilleachair, N., & Ó Catháin, C. (2021). How does the dietary intake of female field-based team sport athletes compare to dietary recommendations for health and performance? A systematic literature review. *Nutrients*, **13**(4), 1235.

Under review: *BMC Public Health* (first author, HEATS protocol); *Performance Nutrition* (first author, second-round review); *Public Health Nutrition* (first author).

FUNDING, SUPERVISION & SERVICE

Funding — Climate Impact Science Research Programme, NRF Singapore & NEA (**CISR-2024-2R-04**): research team member leading the evidence-synthesis workstream for Singapore’s Exertional Heat Vulnerability Index; protocol lead and review guarantor. Contributor to Project HEATS under the NRF CREATE programme.

Supervision — Three PhD candidates and one postdoctoral researcher at NUS: Raagavi Mani (dormitory heat and sleep), Gabriel May Moe Kyaw (nighttime heat thresholds, CRD4202511009764), Jessica Loke (fluid temperature and cooling, CRD420251134107), and Dr Warinthorn Angkanasirikul (heat vulnerability index reporting). Previously supervised ten undergraduate research projects at ATU.

Teaching — Invited presenter, “Which Test, and Why?”, CLIF Foundational Workshop on Statistical Approaches, NUS, August 2026, with a published open-access [statistical test chooser](#). Featured speaker, Project HEATS workshop, Indoor Air 2026. Former module lead, MSc Advanced Performance Nutrition.

Peer review — *Sports Medicine – Open* (2026); *Environmental Science and Pollution Research International* (2025); *Journal of Science in Sport and Exercise* (2021); *Journal of the International Society of Sports Nutrition* (2021).

Service — Lead for Strategic Partnerships, Business Development & External Relations, NUS Medicine Early Career Researchers Office (2026–27). Education Subcommittee Lead, NUS Medicine Postdoctoral Association.

EDUCATION & AWARDS

PhD, Sport, Health and Exercise Sciences — Technological University of the Shannon, Ireland, 2024
Determinants of Dietary Intake in Team Sport Athletes

MSc, Applied Sports Nutrition — St Mary’s University Twickenham, UK, 2019

BSc (Hons), Exercise, Nutrition and Health — Kingston University, UK, 2017

JSPS HOPE Fellow, Japan Society for the Promotion of Science, 2026 — selected to represent NUS Medicine at the 17th JSPS HOPE Meeting with Nobel Laureates, Tsukuba.

Winner, Postgraduate Competition, The Nutrition Society (UK & Ireland), 2024 — invited lecture, International Congress, Belfast.